

Maike L Morrison

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Appointments

2025– **Santa Fe Institute** **Santa Fe, NM**
Omidyar Postdoctoral Research Fellow
Prof. Graham Coop (UC Davis), external mentor

Education

2020–2025 **Stanford University** **Stanford, CA**
PhD in Biology (Ecology & Evolution)
Prof. Noah Rosenberg, advisor
Associate Teaching Certificate

2016–2020 **The University of Texas at Austin** **Austin, TX**
BS in Mathematics, Dean's Scholars Honors Program
Profs. Lauren Ancel Meyers & Mark Kirkpatrick, advisors

Peer-Reviewed Publications

6. **Morrison ML**, Xue KS, Rosenberg NA (2025). Quantifying compositional variability in microbial communities with FAVA. *Proceedings of the National Academy of Sciences*, 122(11), e241321122.
5. **Morrison ML** & Rosenberg NA (2023). Mathematical bounds on Shannon entropy given the abundance of the i th most abundant taxon. *Journal of Mathematical Biology*, 87(5), 76.
4. **Morrison ML**, Alcalá N, Rosenberg NA (2022). FSTRUCT: an F_{ST} -based tool for measuring ancestry variation in inference of population structure. *Molecular Ecology Resources*, 22(7), 2614-2626.
3. Ingle T, **Morrison ML**, Wang X, Mercer T, Karman V, Fox S, Meyers LA (2021). Projecting COVID-19 isolation bed requirements for people experiencing homelessness. *PLOS One*, 16(5), e0251153.
2. Kachuri L*, Francis SS*, **Morrison ML**, Wendt GA, Bossé Y, Cavazos TB, Rashkin SR, Ziv E, Witte JS (2020). The landscape of host genetic factors involved in immune response to common viral infections. *Genome Medicine* 12(93), 1-18.
1. **Morrison M***, Castro LA*, Meyers LA (2020). Conscientious vaccination exemptions in kindergarten to eighth-grade children across Texas schools from 2012 to 2018: A regression analysis. *PLOS Medicine* 17(3), e1003049.

Preprints

6. Klass K, Van Belle S, ..., **Morrison ML**, ..., Teichroeb JA, Greenbaum G (2026). Contrasting effects of forest fragmentation on the genetics and microbiomes of an endangered arboreal primate. *bioRxiv*.
5. **Morrison ML**, Woodhouse A, Swain A (2026). Latitudinal dependence of stability trends in marine plankton in the Cenozoic. *bioRxiv*.
4. Sexton-Oates A*, Mathian E*, Candeli N, ..., **Morrison ML**, ..., Alcalá N, Foll M, Fernandez-Cuesta L (2025). A clinically relevant morpho-molecular classification of lung neuroendocrine tumours. *medRxiv*.
3. **Morrison ML**, Mangé L, Senkin S, Rosenberg NA, Foll M, Fernandez-Cuesta L, Alcalá N (2023). Variability of mutational signatures is a footprint of carcinogens. *medRxiv*.
2. Xue KS, Walton SJ, Goldman DA, **Morrison ML**, ..., Huang KC, Good BH, Relman DA (2023). Prolonged delays in human microbiota transmission after a controlled antibiotic perturbation. *bioRxiv*.

1. Miyagi M, **Morrison M**, Kirkpatrick M (2022). How many ecological niches are defined by the super-abundant marine microbe *Prochlorococcus*? *bioRxiv*.

* These authors contributed equally

Software Packages

FAVA R package: An F_{ST} -based Assessment of Variability over relative Abundances

sigvar R package: Quantify and visualize variability of mutational signatures within and across samples

FSTruct R package: Measure variability in population structure estimates

Research Positions

2023	Visiting Scientist, International Agency for Research on Cancer	Lyon, France
2022	Santa Fe Institute Complex Systems Summer School	Santa Fe, NM
2019	UCSF Amgen Scholars Program: Prof. John Witte	San Francisco, CA
2018	Mathematical Biosciences Institute NSF REU: Prof. Ephraim Hanks	Columbus, OH

Research Fellowships

2025–2028	Santa Fe Institute Omidyar Postdoctoral Research Fellowship
2021–2025	NSF Graduate Research Fellowship Program
2020–2025	Stanford Graduate Fellowship

Teaching

Winter 2025	BIO 192/387: Statistical Methods in Ecology & Evolution <i>Instructor of Record</i> · Designed and taught course on the theory and application of biological diversity statistics.	Stanford University <i>Faculty advisor: Dr. Noah Rosenberg</i>
Fall 2023	BIO 388: CEHG Seminar - Microbiome Ecology & Evolution <i>Instructor of Record</i> · Co-organized Fall 2023 Computational, Evolutionary, & Human Genomics seminar. · Facilitated accompanying class, including lecturing on statistical methods for biology.	Stanford Univ. <i>Co-taught with Dr. Noah Rosenberg</i>
Winter 2022	BIO 187: Mathematical Population Biology <i>Graduate Teaching Assistant</i> · Prepared and presented lectures on mathematical bounds on F_{ST} and Shannon entropy.	Stanford University <i>for Dr. Noah Rosenberg</i>
Winter 2021	BIO 85: Evolution <i>Graduate Teaching Assistant</i> · Wrote & graded assessments and organized discussion sections for introductory online course.	Stanford University <i>for Drs. Dmitri Petrov & Molly Schumer</i>

Academic Service

2025–2026	Co-Organizer, IF/THEN Seminar Series	Santa Fe Institute
2024–2025	Member, CEHG Symposium Organizing Committee <i>Center for Computational, Evolutionary, and Human Genomics (CEHG)</i>	Stanford University
2023–2024	Student Member, Ecology & Evolution PhD Admissions Committee	Stanford University

2021–2022	Co-Organizer, Eco-Evo Lunch	Stanford University
2021	Co-Organizer, Biology Pre-Orientation Program (BioPOP)	Stanford University
2016–2020	Member, Dean’s Scholars Council (Chair 2018–2019)	University of Texas at Austin

Peer review activity: *Genetics, Methods in Ecology and Evolution, Molecular Biology & Evolution, Proceedings of the National Academy of Sciences, Theoretical Population Biology*

Mentorship: Two undergraduate researchers (Stanford & UC Davis), two undergraduates through the Stanford Women’s Community Center, several prospective/incoming Stanford graduate students.

Awards

2025	Stanford Biology Department Samuel Karlin Thesis Prize in Mathematical Biology
2025	Stanford University Centennial Teaching Assistant Award
2025	Outstanding Poster, Society for Modeling & Theory in Population Biology Meeting
2024	Outstanding Oral Presentation, Society for Modeling & Theory in Population Biology Meeting
2022	Graduate Student Poster Award, Genetics Society of America PEQG meeting
2022	Stanford Biology Department Excellence in Teaching Award
2020	Dean’s Honored Graduate: University of Texas College of Natural Sciences
2020	Graduate Distinction in Service & Leadership: University of Texas College of Natural Sciences
2020	Graduate Distinction in Research: University of Texas College of Natural Sciences
2020	George H. Mitchell Award for Outstanding Undergraduate Research
2020	UT Austin Mathematics Special Departmental Honors
2019	Alan Kaylor Cline Dean’s Scholars Scholarship
2019	Phi Beta Kappa, junior year election
2019	Undergraduate Diversity Award for the 2019 Evolution Meeting
2019	Barry M. Goldwater Scholarship

Conference Presentations

Selected Oral Presentations

June 2026	FORAMS 2026	Amherst, MA
	“Latitudinal dependence of stability trends in Cenozoic marine plankton”	
June 2026	Population, Evolutionary, & Quantitative Genomics Conference	Pacific Grove, CA
	“A general F_{ST} framework reveals the variability of rare versus common alleles”	
Nov 2024	Bay Area Population Genetics Conference	Berkeley, CA
	“Quantifying compositional variability in microbial communities”	
May 2024	Modeling & Theory in Pop. Bio. (Banff Int. Research Station)	Banff, AB
	“Quantifying the stability of microbiomes and the timescale of antibiotic perturbation”	
May 2024	American Mathematical Society Spring West. Sect. Meeting	San Francisco, CA
	“Quantifying compositional variability in microbial communities”	
	<i>Invited talk</i>	

Aug 2023 **Ecological Society of America Annual Meeting** Portland, OR
 “Quantifying compositional variability in microbial communities”

Selected Poster Presentations

June 2025 **Society for Modeling & Theory in Population Biology** Chicago, IL
 “A general F_{ST} framework reveals the variability of rare versus common alleles”

July 2024 **Society for Molecular Biology & Evolution** Puerto Vallarta, Mexico
 “Quantifying the stability of microbiomes and the timescale of antibiotic perturbation with FAVA”

Mar 2024 **The Allied Genetics Conference** Washington, DC
 “Quantifying compositional variability in microbial communities”

Dec 2022 **The Human Microbiome: Ecology and Evolution (Keystone Symposia)** Banff, AB
 “A diversity statistic for quantifying microbiome stability”

June 2022 **Population, Evolutionary, & Quantitative Genetics Conference** Pacific Grove, CA
 “FSTruct: An F_{ST} -based tool for quantifying ancestry variability”

June 2019 **Evolution Meeting** Providence, RI
 “How many ecotypes are comprised by the world’s most abundant photosynthetic organism?”

Selected Workshops

June 2026 **FORAMS 2026** Amherst, MA
 “Triton database workshop: Statistical methods for spatiotemporal variation”

June 2026 **Society for Modeling & Theory in Population Biology** Providence, RI
 “Parallel models and statistics across ecology and evolution”

Selected News/Media

Aug 2026 **Guest column:** “Federal funding brought me from Canyon H.S. to cancer research — cuts threaten the next generation,” for the New Braunfels, TX *Herald-Zeitung*.

Mar 2020 **Live television interview:** Discussed Texas vaccination exemptions with Amanda Salinas on FOX 7’s *Good Day Austin*, an Austin-area morning show.

Mar 2020 **Media coverage:** “Rich White Texans Are Running With Anti-Vax Misinformation,” by Julia Ries, *VICE*.

Mar 2020 **Media coverage:** “White, College-Educated, Wealthier Parents Less Likely to Vaccinate their Children, Texas Study Reveals,” by Kashmira Gander, *Newsweek*.